10.1 Parametrizations of Plane Curves

Instead of describing a curve as the graph of a function, think of it as a path of a moving particle whose position is changing over time.

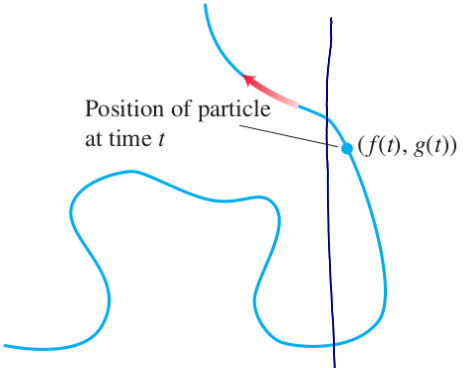


FIGURE 10.1 The curve or path traced by a particle moving in the xy-plane is not always the graph of a function or single equation.

- The path fails the vertical line test
  ⇒ cannot described as a function of x.
- Instead, we can describe as
  $(x, y) = (f(t), g(t))$ for some continuous function.
- t = usually denotes time.

Definition: If x and y are given as functions $x = f(t),\ y = g(t)$ over an interval I of t-values, then the set of points $(x, y) = (f(t), g(t))$ defined by these equations is a **parametric curve**. The equations are **parametric equations** for the curve.

t = parameter of the curve

I = parameter interval e.g. $a \le t \le b \Rightarrow I = [a, b]$.

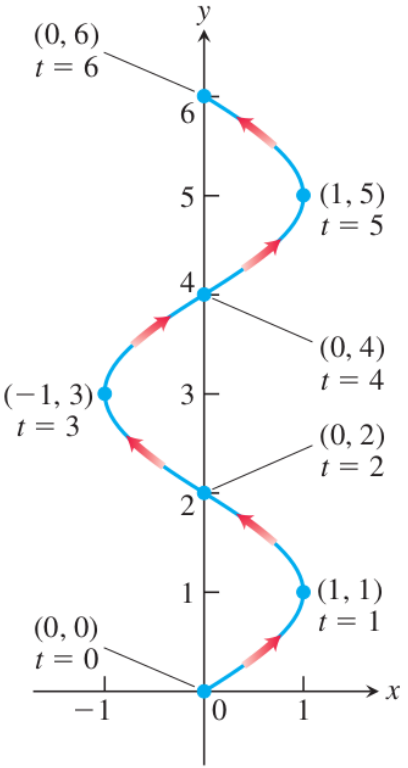
If $I = [a, b]$, then $(f(a), g(a))$ = initial point & $(f(b), g(b))$ = terminal point.

Example: Sketch the curve defined by the parametric equations

$$x = \sin \pi t/2, \quad y = t, \quad 0 \le t \le 6.$$

TABLE 10.1 Values of $x = \sin \pi t/2$ and $y = t$ for selected values of t.

t	x	y
0	0	0
1	1	1
2	0	2
3	−1	3
4	0	4
5	1	5
6	0	6



Example: Sketch the curve defined by the parametric equations

$$x = t^2, \quad y = t + 1, \quad -\infty < t < \infty.$$

$t = y - 1 \Rightarrow x = (y-1)^2 = y^2 - 2y + 1$

TABLE 10.2 Values of $x = t^2$ and $y = t + 1$ for selected values of t.

t	x	y
-3	9	-2
-2	4	-1
-1	1	0
0	0	1
1	1	2
2	4	3
3	9	4

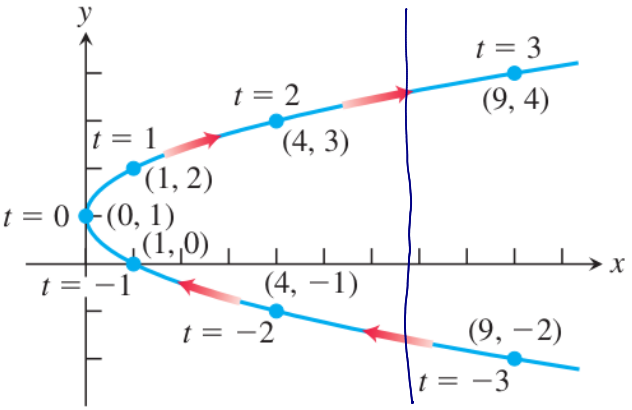


FIGURE 10.3 The curve given by the parametric equations $x = t^2$ and $y = t + 1$ (Example 2).

↳ x is a function of y, $y \in \mathbb{R}$.

Example: Graph the parametric curve

$$x = a\cos t,\quad y = a\sin t,\quad 0 \le t \le 2\pi$$

$a > 0$

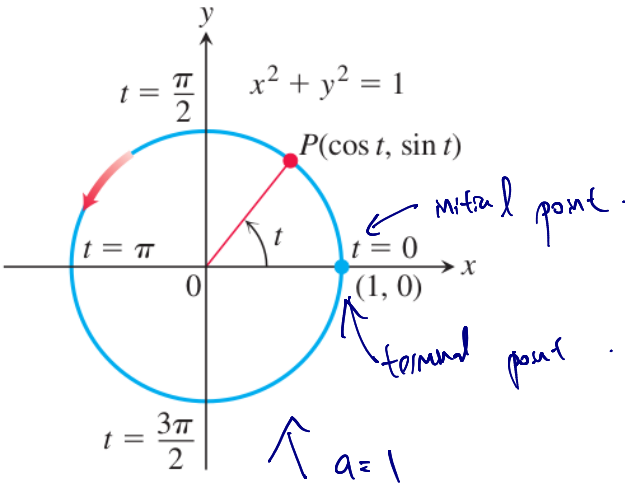


FIGURE 10.4 The equations $x = \cos t$ and $y = \sin t$ describe motion on the circle $x^2 + y^2 = 1$. The arrow shows the direction of increasing t (Example 3).

$x^2 + y^2 = a^2\cos^2 t + a^2\sin^2 t$

$= a^2(\cos^2 t + \sin^2 t)$

$= a^2$.

Circle of radius a centered @ origin.

Example: The position $P(x, y)$ of a particle moving in the xy-plane is given by the equations and parameter interval

$$x = \sqrt{t}, \quad y = t, \quad t \geq 0.$$

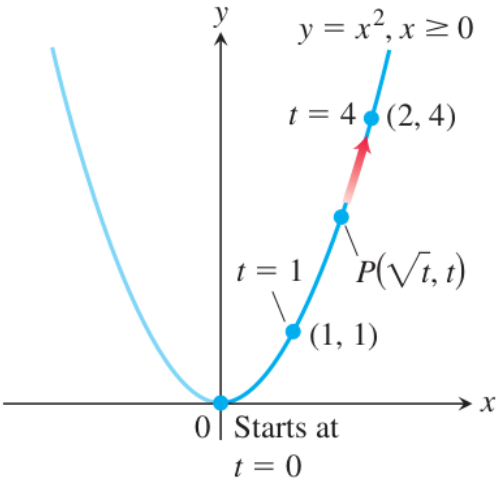


FIGURE 10.5 The equations $x = \sqrt{t}$ and $y = t$ and the interval $t \geq 0$ describe the path of a particle that traces the right-hand half of the parabola $y = x^2$ (Example 4).

$y = t = (\sqrt{t})^2 = x^2$

This is an equation of a parabola.

Since $x = \sqrt{t} \geq 0$

only the right half of

the parabola $y = x^2$.

Example: Find a parametrization for the line through the point (a, b) having slope m.

line equation : $y - b = m(x - a)$.

$\Rightarrow y = m(x-a) + b$.

If $t = x - a$ we get

$x = t + a$.

$y = m(x-a) + b = mt + b$.

$-\infty < t < \infty$

$x = t$

$y = m(t-a) + b$

$t \in \mathbb{R}$

Both parametrizations describe the same line.

Example: Sketch and identify the path traced by the point $P(x, y)$ if

$$x = t + \frac{1}{t}, \quad y = t - \frac{1}{t}, \quad t > 0.$$

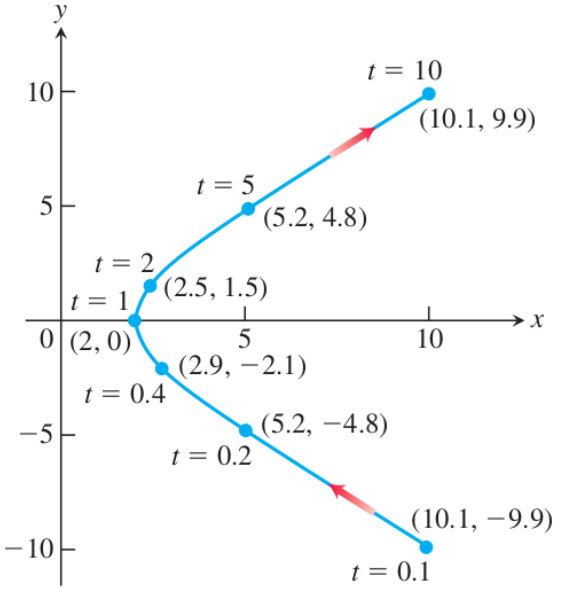


FIGURE 10.7 The curve for $x = t + (1/t)$, $y = t - (1/t)$, $t > 0$ in Example 7. (The part shown is for $0.1 \le t \le 10$.)

eliminate t

$x - y = \frac{2}{t}$

$x + y = 2t$

$(x-y)(x+y) = 4$

$\Rightarrow x^2 - y^2 = 4$.

An equation of a hyperbola.

Other parametrizations of $x^2 - y^2 = 4,\ x > 0$.

- $x = \sqrt{4+t^2}$, $y = t$, $-\infty < t < \infty$

$$x^2 - y^2 = \left(\sqrt{4+t^2}\right)^2 - t^2 = 4 \quad .$$

- $x = 2\sec t$ $\quad y = 2\tan t$ $\quad -\frac{\pi}{2} < t < \frac{\pi}{2}$.

$\Rightarrow x > 0$

$$x^2 - y^2 = 4\sec^2 t - 4\tan^2 t$$
$$= 4\left(\sec^2 t - \tan^2 t\right)$$
$$= 4 .$$